Expressive Semantic Reasoning in Context-Aware Systems

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Abstract. Context-Aware Systems are those that can adapt their operations to the current context in which they are developed, without the need for explicit user intervention. In the current work, we propose the use of Expressive Description Logic, to model reasoning problems of Context-Aware Systems.

Keywords: Context-Aware Systems, Semantic Reasoning, Description Logic.

1 Introduction

The Semantic Web is a set of technologies, standards that serve us to transmit, store data on the internet, in it we find specific technologies such as RDF, RDF SCHEMA, SPARQL, OWL among others[9]. In relation to Web Ontology Language in specific is a family of languages for the representation of knowledge[11], it has a great importance because it is a more expressive ontological language based on the Description Logic, OWL consists of three species relevant [8], namely OWL Lite, OWL 1 and OWL 2 which are intended to be layered according to increasing expressiveness, therefore, we examine in this document the OWL language as it is and the properties in the construction of OWL models [10].

The Context-Aware System is a system in general that is capable of taking information from its environment, which has to adapt and react according to its context, for this type of systems there is a degree of complexity in its development or construction inherent to the development[5], for example, it is common for a system to present faults and not foresee to reduce this type of problems, the reasoning modeling is carried out according to the tasks of each system, achieving more reliable and robust systems.That is why it is important to model and reason around Context-Aware System [8] [4].

OWL 2 is one of the most expressive languages for modeling knowledge in general has some limitations such as being able to model spatial and temporal issues at the same time [3], the complexity of automatic reasoning has generated the need to investigate and create new contributions in the representation of knowledge. The usefulness of the reasoning and modeling tasks of Context-Aware System helps us to build more robust and secure systems [12].

The construction of a good formalism that model contextual information reduces the complexity of the applications of a Context-Aware System and improves the maintenance of a system [5],therefore the existence of a well-designed model of contextual information facilitates the development and operation of systems. On the other hand, it is important that the system meets the tasks of reasoning in general by giving an example, one of the tasks is consistency, its function is to ensure that there are no contradictions, our system must conclude some fact or the opposite [7].

Based on the expressive power of the Description Logics and the different developed reasoning algorithms that solve the satisfiability and validity of a model, the use of Expressive Description Logics is proposed, since they are formal logic, they allow reasoning on a defined knowledge base, as a basis for solve Context-Aware Systems. In particular, identifying the different types of reasoning problems used to demonstrate the consistency between different language concepts.

It is important to distinguish that our task is not to build a Context-Aware System. The main objective of this article is model reasoning problems in Context-Aware Systems using a Description Logic for model context more broadly, definitely it will be a very expressive system. We propose a solution for resolve problems of Context-Aware System which involves expressive Description Logic that uses arithmetic operators that support in reducing the limitations of OWL 2.

2 Interpretation of expressive description logic for trees

For this research work, several studies have been carried out that encompass many factors, be borne in mind the advantages, contributions and identifying the areas that still need to be covered.

The researches based on logic has allowed to have a more wide panorama, for example such as describes Blackburn [6] [1] "a Description Logic are a family of knowledge representation languages which can be used to represent the terminological knowledge of an application domain in a structured and formally well-understood way", they base their expressivity on their logical operators where computational complexity is related with the amount of resources an algorithm uses.

Since our main motivation is to be able to represent Context-Aware Systems, a problem that is detected in the original Description Logics is not being able to decide if one concept is a sub-concept of another, so that in order to solve this problem we base our research on a syntactic variant of the language of Description Logics that is equipped with operators like negation, conjunction, regular roles, inverse roles, nominals and qualified number restrictions as Barcenas showed[3].

2.1 Syntax and Semantics

In this section it's exposed an alphabet assuming it's formalisms, therefore, it's necessary to specify a finite set of roles that are constituted in the following way: $\{Ch, Rsib, Par, Lsib\}$

Definition 1 (Syntax). The concepts descriptions are defined by the following grammar.

$$C := A \mid \neg C \mid C \sqcup C \mid \exists R.C \mid C - C > k \mid \mu x.C$$
$$R := Ch \mid Rsib \mid Par \mid Lsib$$

The concepts descriptions are defined as follows C represents a concept can be basic or compound, A belongs to a set of concepts names, $\neg C$ represents the denial of a concept, $C_1 \sqcup C_2$ represents the disjunction of two concepts, $\exists R.C$ is represented with the operator $\exists$ analogously R is a finite set of roles and C represents a concept, $C - C > k$ which is interpreted as the children nodes that meet a numerical constraint $>k$ which represents a positive natural number, $\mu x.C$ is represented by a variable and a concept, defined as the fixed minimum points, whose interpretation is defined as a finite recursion, it's important to remember that the concepts descriptions are interpreted on tree structures as node subsets.

Also, it's important to consider the special syntax such as $C_1 \sqcap C_2 := \neg(\neg C_1 \sqcup \neg C_2)$ what does it mean the intersection of concepts, $\top := C \sqcap \neg C$ it allows us to represent a the entire set of concepts nodes, $\bot := \neg \top$ analogously, denote a tautology of concepts, $\forall R.C := \neg \exists R.\neg C$ it's the representation of nodes where all the nodes related through R are found in the interpretation of the concept C, finally $vx..C := \mu x.\neg C[\neg x/x]$ by a set of many variables as a greatest fixed point.

It is necessary to define a tree interpretation in the following way $\mathcal{I} = \{\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}}\}$ first $\Delta^{\mathcal{I}}$ the domain is defined as finite non-empty set of nodes, followed by an interpretation function $\mathcal{I}$ where the domain are going to be the concept names and they will give us as a result their nodes, for example A a set $(A)^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$ and each role name R is assigned a set of pairs of nodes $(R)^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}}$ to be more precise, each role is defined as follows $(Ch)^{\mathcal{I}}$, $(Rsib)^{\mathcal{I}}$, $(Par)^{\mathcal{I}}$ and $(Lsib)^{\mathcal{I}}$ finally, it is necessary to define variables that are interpreted by means of a valuation function ρ that encompass domain subsets $\rho(x) \subseteq \Delta^{\mathcal{I}}$.

Definition 2 (Semantics). Given a structure of a set or domain and a binary relation that will assign to each concept name a node, in terms of a tree interpretation. $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ Therefore, it's defined as a valuation ρ on $\mathcal{I}$ thats allow to represent the concept representation in the following way:

$$\begin{aligned}
(x)^{\mathcal{I}}_{\rho} =& \rho(x) \subseteq \Delta^{\mathcal{I}} \\
(A)^{\mathcal{I}}_{\rho} =& (A)^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}} \\
(\neg C)^{\mathcal{I}}_{\rho} =& \Delta^{\mathcal{I}} \backslash (C)^{\mathcal{I}}_{\rho} \\
(C_1 \sqcup C_2)^{\mathcal{I}}_{\rho} =& (C_1)^{\mathcal{I}}_{\rho} \cup (C_2)^{\mathcal{I}}_{\rho} \\
(\exists R.C)^{\mathcal{I}}_{\rho} =& \{s \in \Delta^{\mathcal{I}} \mid (s, s') \in (R)^{\mathcal{I}}, s' \in (C)^{\mathcal{I}}_{\rho}\} \\
(C_1 - C_2 > k)^{\mathcal{I}}_{\rho} =& \{s \in \Delta^{\mathcal{I}} \mid \\
& \mid \{s' \mid (s, s') \in (Ch)^{\mathcal{I}}, s' \in (C_1)^{\mathcal{I}}_{\rho}\} \mid - \\
& \mid \{s'' \mid (s, s'') \in (Ch)^{\mathcal{I}}, s'' \in (C_2)^{\mathcal{I}}_{\rho}\} \mid > k\} \\
(\mu x.C)^{\mathcal{I}}_{\rho} =& \bigcap \{\varepsilon \subseteq \Delta^{\mathcal{I}} \mid (C)^{\mathcal{I}}_{\rho[x/\varepsilon]} \subseteq \varepsilon\}
\end{aligned}$$

Given an interpretation of tree $\mathcal{I}$ that satisfies a concept description C if the interpretation is different from the empty set $\rho, (C)^{\mathcal{I}}_{\rho} \neq \emptyset$ it is possible to say that it is an interpretation of model C.

For example suppose we have a set of concepts descriptions, according to the language described, where the children nodes have some relation to the descriptions of concepts. we can represent the following sentence in formal language:

$$\exists Ch.A_1 \sqcup \exists Ch.A_3$$

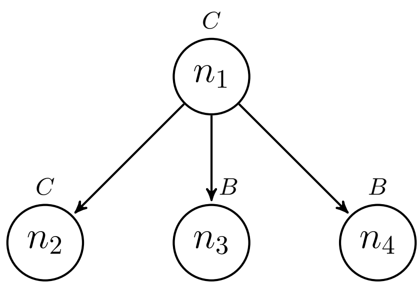

Fig. 1. Tree structure that satisfies the formula 1.1: $(B - A) > 0$.

Example 1.1 A concept description is defined where the node $\{n_1\}$ satisfies the formula.

$$(B - A) > 0$$

According to the figure 1 the tree model is the example (a) defined in a tree interpretation $\mathcal{I} = \{\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}}\}$ his domain is defined as $\Delta^{\mathcal{I}} := \{n_1, n_2, n_3, n_4\}$ the names of concepts are represented as follows $(A)^{\mathcal{I}} = \{n_1, n_2\}$, $(B)^{\mathcal{I}} = \{n_3, n_4\}$ and their roles are $(Ch)^{\mathcal{I}} = \{(n_1, n_2), (n_1, n_3), (n_1, n_4)\}$, followed by parents $(Par)^{\mathcal{I}} = \{(n_2, n_1), (n_3, n_1), (n_4, n_1)\}$, and the right siblings $(Rsib)^{\mathcal{I}} = \{(n_2, n_3), (n_3, n_4), \}$ finally, by the left siblings $(Lsib)^{\mathcal{I}} = \{(n_4, n_3), (n_3, n_2)\}$.

$$(B - A) > 0$$

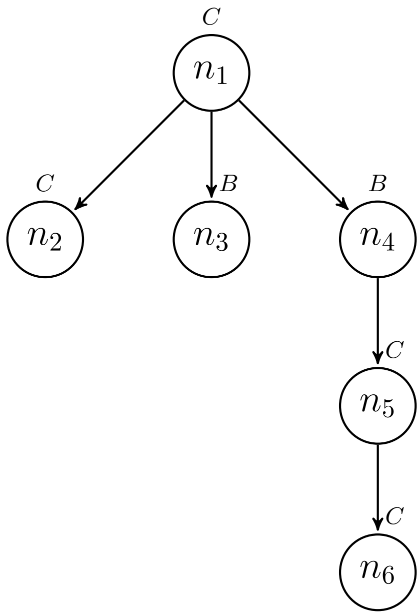

Fig. 2. Tree structure that satisfies the formula 1.2: $\mu x.C \sqcup \exists Ch.x$.

Example 1.2 A concept description is defined where the node $\{n_5\}$ satisfies the formula.

$$\mu x.C \sqcup \exists Ch.x$$

Another example, given the concept description where the node $\{n_5\}$ satisfies the formula. According to the figure 2 the tree model is the example (b) defined in a tree interpretation $\mathcal{I} = \{\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}}\}$ where the domain is defined as the set of nodes $\Delta^{\mathcal{I}} := \{n_1, n_2, n_3, n_4, n_5, n_6\}$ assigned to the concepts names $(A)^{\mathcal{I}} = \{n_1, n_2\}$, $(B)^{\mathcal{I}} = \{n_3\}$ and $(C)^{\mathcal{I}} = \{n_4, n_5, n_6\}$ the concepts names are defined, firstly the children are defined $(Ch)^{\mathcal{I}} = \{(n_1, n_2), (n_1, n_3), (n_1, n_4), (n_4, n_5), (n_5, n_6)\}$, in next place we define the parents $(Par)^{\mathcal{I}} = \{(n_2, n_1), (n_3, n_1), (n_4, n_1), (n_5, n_4), (n_6, n_5)\}$, followed by the right siblings $(Rsib)^{\mathcal{I}} = \{(n_2, n_3), (n_3, n_4), \}$ and finally, by the left siblings $(Lsib)^{\mathcal{I}} = \{(n_4, n_3), (n_3, n_2), \}$.

$$\mu x.C \sqcup \exists Ch.x$$

2.2 Nominals

Through the operators we have, it's possible to ensure that a formula is true in one part of the model, therefore, a nominal is defined as follows:

$$I := \varphi \sqcap siblings(\neg\varphi) \sqcap ancestors(\neg\varphi) \sqcap descendants(\neg\varphi)$$

Where *I* is our nominal φ a concept name can be true anywhere but in all of its siblings, it's not true and in all it's ancestors also and in all it's descendants it also means that it will be true in a single node.

$$siblings(C) := \forall Rsib.C \sqcap \forall Lsib.C$$
$$ancestors(C) := \forall Par.\mu x.C \sqcap siblings(C) \sqcap \forall Par.x$$
$$descendants(C) := \forall Ch.\mu x.C \sqcap \forall Ch.x$$

We deduct the nominal will be true in one place inside the model, unlike the normal propositions that could be true in different places inside the model.
As the term itself suggests, it is possible to express the following proposition.
Proposition 1 "If the interpretation of the concept with respect to the tree is different from the empty set then we say that this tree is a model for the concept" $\mid (I)^{\mathcal{I}}_{\rho} \mid = 1$ or $\mid (I)^{\mathcal{I}}_{\rho} \mid = 0$.

2.3 Terminology

A TBox is defined as terminological axioms that are used to make statements about how the concepts are kept in relation. In the construction of a model if a new concept is defined, it is transcendental to know if it is consistent or not in relation to the TBox. We know this property as satisfiability [1,2,3].

It is necessary to formalize the following properties, first suppose that T is defined as a TBox, and C_1 and C_2 are concepts. For example:

$$C_1 \sqsubseteq C_2$$

Where C_1 and C_2 are concepts, so C_1 is subsumed by C_2 with respect to T if for every interpreted of model $\mathcal{I}$ it's a model of T, $C_1 \sqsubseteq C_2$, if and only if $(C_1)^I_\rho \subseteq C_2)^I_\rho$ for any valuation ρ in $\mathcal{I}$.

A TBox T implies a concept description of C_2 where there is a model that satisfies T if and only if it exists for any interpretation of tree $\mathcal{I}$ and any valuation of ρ. For example.

$$(C_T \sqcup C)^I_\rho \neq 0$$

The use of TBox in a Context-Aware Systems is important since an activity can be inferred to know if it is consistent or contradictory with TBox. In other words, this property will allow us to know if a concept of our given model is satisfactory or not.

2.4 Worlds Description

On the other hand, an assertion determines the instance of a concept in an individual and a role in a pair of individuals[1,2,3]. For example

Abox: Describe Individuals
$D(Ch)$
$Rsib, LSib(C, Ch)$
$Par(C)$

The use of ABox in a Context-Aware System is fundamental given that the problems of satisfiability, subsumption concepts, and consistency are identified, these being the basic types of inference of the Description Logic, therefore when developing an algorithm, we will make sure to solve the problems for a Context-Aware System.

3 Conclusions

The use of Expressive Descriptive Logics gives the possibility of modeling reasoning problems with specific operators that are provided by the same language, therefore, it is essential to develop an algorithm associated with Expressive Description Logics, adding the different language operators to obtain different expressivities, which allow to solve problems of Context-Aware Systems. and automatic efficient reasoning.

A further research is to develop efficient reasoning algorithms that allow to be more expressive, according to the expressive Description Logics we believe that different language operators will allow us to model complex reasoning tasks in Context-Aware Systems. For example, if the objetive is to resolve the consistency, a problem will be characterized in terms of satisfability and it will be ensured in carrying out different experiments that allow applying said conclusions in a model given.

References

1. Baader, F., Calvanese, D., McGuinness, D.L., Nardi, D., Patel-Schneider, P.F. (eds.): The Description Logic Handbook: Theory, Implementation, and Applications. Cambridge University Press (2003)

2. Baader, F., Hollunder, B., Nebel, B., Profitlich, H., Franconi, E.: An empirical analysis of optimization techniques for terminological representation systems, or making KRIS get a move on. In: Proceedings of the 3rd International Conference on Principles of Knowledge Representation and Reasoning (KR'92). Cambridge, MA, October 25-29, 1992. pp. 270–281 (1992)
3. Barcenas, E., Molero, G., Sanchez, G., Benitez-Guerrero, E., Mezura-Godoy, C.: Reasoning on expressive description logics with arithmetic constraints. In: 2016 International Conference on Electronics, Communications and Computers (CONIELECOMP). pp. 180–185 (Feb 2016)
4. Benítez-Guerrero, E.: Context-aware mobile information systems: Data management issues and opportunities. In: Arabnia, H.R., Hashemi, R.R., Vert, G., Chennamaneni, A., Solo, A.M.G. (eds.) International Conference on Information & Knowledge Engineering, IKE. pp. 127–133 (2010)
5. Bettini, C., Brdiczka, O., Henricksen, K., Indulska, J., Nicklas, D., Ranganathan, A., Riboni, D.: A survey of context modelling and reasoning techniques. Pervasive Mob. Comput. 6(2), 161–180 (Apr 2010), `http://dx.doi.org/10.1016/j.pmcj.2009.06.002`
6. Blackburn, P., van Benthem, J.F., Wolter, F.: Handbook of modal logic, vol. 3. Elsevier (2006)
7. Limón, Y., Bárcenas, E.: On the consistency of context-aware systems. Research in Computing Science 22, 1–11 (2018)
8. Limón, Y., Bárcenas, E., Benítez-Guerrero, E., Mezura-Godoy, C.: Towards a reasoning model for context-aware systems: Modal logic and the tree model property. Research in Computing Science 99, 9–18 (2015)
9. Maarala, A.I., Su, X., Riekki, J.: Semantic reasoning for context-aware internet of things applications. IEEE Internet of Things Journal 4(2), 461–473 (November 2017)
10. Riboni, D., Bettini, C.: Owl 2 modeling and reasoning with complex human activities. Pervasive Mob. Comput. 7(3), 379–395 (Jun 2011), `http://dx.doi.org/10.1016/j.pmcj.2011.02.001`
11. W3C: Web Ontology Language (OWL) (2011), `https://www.w3.org/OWL/`
12. Wang, J., Li, B.: Study of semantic reasoning based on ontology description logic. In: 2016 IEEE Advanced Information Management, Communicates, Electronic and Automation Control Conference (IMCEC). pp. 1869–1872 (Oct 2016)